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MONGO -DB CRUD OPERATION:

```
import com.mongodb.client.FindIterable;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import
com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document; public class Test2
{
public static void main(String[] args) {
// Connect to MongoDB
MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");
// Access Database and Collection
MongoDatabase database = mongoClient.getDatabase("CollegeDB");
MongoCollection<Document> collection = database.getCollection("Students");
// =====
// CREATE (Insert 5 Students)
// =====
collection.insertOne(new Document("id", 101)
.append("name", "Aarav")
.append("department", "CSE")
```

```
.append("cgpa", 8.5));
Document("id", 102)
collection.insertOne(new
.append("name", "Priya")
.append("department", "IT")
.append("cgpa", 9.1));
Document("id", 103)
collection.insertOne(new
.append("name", "Rahul")
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.append("department", "ECE")
.append("cgpa", 8.0));
Document("id", 104)
collection.insertOne(new
.append("name", "Sneha")
.append("department", "ME")
.append("cgpa", 8.8));
collection.insertOne(new Document("id", 105)
.append("name", "Vikram")
.append("department", "CIVIL")
.append("cgpa", 7.9));
System.out.println("5 Documents Inserted Successfully");

// =====
// READ (Display All Students)
// =====
```

```

System.out.println("\nStudent Records:");

FindIterable<Document> students = collection.find();

(Document doc : students) {

System.out.println(doc.toJson());

}

// =====

// UPDATE

// Update CGPA of Student ID 103

// =====

collection.updateOne(

Filters.eq("id", 103),

Updates.set("cgpa", 9.3)

);

for

System.out.println("\nStudent ID 103 Updated Successfully");

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// =====

// READ Updated Record

// =====

Document updatedStudent =

collection.find(Filters.eq("id", 103)).first();

System.out.println("\nUpdated Record:");

(updatedStudent != null) {

if

System.out.println(updatedStudent.toJson());

```

```

}

// =====

// DELETE

// Delete Student ID 105

// =====

collection.deleteOne(Filters.eq("id", 105));

System.out.println("\nStudent ID 105 Deleted Successfully");

// =====

// DISPLAY FINAL RECORDS

// =====

System.out.println("\nFinal Student Records:");

for (Document doc : collection.find()) {

System.out.println(doc.toJson());

}

// Close Connection

mongoClient.close();

}

```

For Many:

```

package mongodb2; import

com.mongodb.client.FindIterable; import

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com.mongodb.client.MongoClient; import

com.mongodb.client.MongoClients; import

com.mongodb.client.MongoCollection;

import com.mongodb.client.MongoDatabase;

```

```
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import java.util.Arrays; import
org.bson.Document; public class Test2 {
public static void main(String[] args) {
// Connect to MongoDB
MongoClient mongoClient =
MongoClients.create("mongodb://localhost:27017");
// Access Database and Collection
MongoDatabase database = mongoClient.getDatabase("CollegeDB");
MongoCollection<Document> collection = database.getCollection("Students")
// CREATE (Insert 5 Students)
collection.insertMany(Arrays.asList(
new Document("id", 101)
.append("name", "Aarav")
.append("department", "CSE")
.append("cgpa", 8.5),
new Document("id", 102)
.append("name", "Priya")
.append("department", "IT")
.append("cgpa", 9.1),
Document("id", 103)
new
.append("name", "Rahul")
.append("department", "CSE")
.append("cgpa", 8.0),
```

```
new
Document("id", 104)
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.append("name", "Sneha")
.append("department", "ME")
.append("cgpa", 8.8),
Document("id", 105)
new
.append("name", "Vikram")
.append("department", "CSE")
.append("cgpa", 7.9));
System.out.println("5 Documents Inserted Successfully");
// READ (Display All Students)
System.out.println("\nStudent Records:");
for (Document doc : collection.find()) {
System.out.println(doc.toJson());
}
// UPDATE
collection.updateMany(
Filters.eq("department", "CSE"),
Updates.set("department", "Computer Science")
);
System.out.println("\nMultiple Documents Updated Successfully");
READ Updated Record
System.out.println("\nRecords After Update:");
```

```
for (Document doc : collection.find()) {  
    System.out.println(doc.toJson());  
}  
  
// Delete Many  
collection.deleteMany(  
    Filters.eq("department", "Computer Science")  
);  
  
System.out.println("\nMultiple Documents Deleted Successfully");  
  
//  
  
// DISPLAY FINAL RECORDS  
  
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System.out.println("\nFinal Student Records:");  
for (Document doc : collection.find()) {  
    System.out.println(doc.toJson());  
}  
  
// Close Connection  
mongoClient.close();  
}  
}
```